

REVIEW

Insulator defect detection with deep learning: A survey

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Funding information

Scientific Research Program Funded by Shaanxi Provincial Education Department, Grant/Award Number: 21JK0661; Key Research and Development Projects in Shaanxi Province, Grant/Award Number: 2021GY-306; Key R&D plan of Shaanxi, Grant/Award Numbers: 2021GY-320, 2020ZDLGY09-10; Natural Science Basis Research Plan in Shaanxi Province of China, Grant/Award Numbers: 2022JQ-568, 2022JQ-696; Fundamental Research Funds for the Central Universities, Grant/Award Number: XJS221502

Abstract

With the improvement of smart grid, utilizing unmanned aerial vehicles (UAV) to detect the operation status of insulators has attracted widespread attention. The insulator defects can lead to serious power loss, damage the service life of power lines, and even result in power outages in serious cases. The small-scale object, complex background, and limited-number collected data make insulator defect still a challenging problem. Benefitted by the advances in deep learning, deep learning-based insulator defects have achieved great progress in recent years. In the paper, the authors present a novel systematic survey of these advances, where further analysis about different processing stages methods is introduced: (i) insulator processing stage methods exploit the specific image pre-processing algorithm for data augmentation and low-level vision information extraction; (ii) defect detection stage model can locate and classify diagnosis fault with different task targets, like sequential task strategy and multi-task strategy. In addition, the authors also review publicly available benchmark and datasets. The future research direction and open problem are discussed to promote the development of the community.

1 | INTRODUCTION

Electric power production is the central nerve and power source of contemporary social and economic operation, which also becomes the foundation to ensure normal social production and residents' daily life. As an important part of the power transmission process, transmission lines not only ensure the safe state of the power system, but also ensure the stable development of the national economy. As a significant part of high-voltage transmission lines, insulators play an important role in electrical insulation and mechanical support. Nowadays, outdoor support insulators on high-voltage transmission lines operating in outdoor open-air conditions are affected by electrical, mechanical,

and environmental factors, making them a weak part of the power transmission system. These potential risks would lead to transmission interruption and threaten the security of the power system. Related research [1, 2, 52] indicates that 81.3% of power accidents in power transmission systems are caused by insulator defects. Additionally, researchers [49, 50, 53] find that the cyber-attacks and insulation defects classification also play an important role in online monitoring of the Internet of Things architecture in smart grid recently.

As a key part of power inspection, insulator detection plays an important role to ensure the reliable power supply of transmission lines. The traditional power detection task usually relies on manual patrol, which is dangerous, labour-intensive, and

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FIGURE 1 The mode of overhead power line inspection. The traditional manual patrol is shown on the left and the advanced UAV patrol is shown on the right. UAV, unmanned aerial vehicles.

costly. In recent years, with the development of unmanned aerial vehicles (UAV) and image processing technology, the UAV patrol strategy has gradually replaced the traditional manual patrol strategy, as shown in Figure 1. Compared with the latter, the former has obvious advantages such as lower cost, higher efficiency, and safer operation.

The insulator strings are located near the pylons of the transmission lines and are inclined due to the effect of gravity. Nowadays, the transmission lines pass through various landforms (forests, towns, rivers, lakes, deserts etc.) leading to the complicated background of aerial images captured by UAV. Therefore, these captured aerial insulator images often exist inevitable image quality difficulties: occlusion, blurring, and object interference. In addition, insulators are made of different materials (glass, ceramic, and composite materials) and different shapes due to different voltage levels. The diverse insulator appearance properties would also make it difficult to detect insulator defects in real-world scenarios.

In the past decade, computer vision-based methods have attracted the increasing interest of researchers working on UAV-based insulator defect detection systems. Traditional computer vision-based methods [24–28] attempt to conceive a universal rule to separate the insulators and defects from the background. Specifically, they consider the colours, shapes, texture, and contours as information cues for insulators and defects detection. Thus, some typical methods such as colour threshold segmentation (Zhai et al. [26]), texture feature (Wu et al. [24]), and contour detection (Zhai et al. [25]) have been widely investigated to segment the positive region containing the insulators and defects from aerial images. However, those approaches show limited robustness since insulators and faults have plenty of variations, and their backgrounds are complicated. Although achieving many improvements, traditional computer vision-based methods still suffer from low reliability and cannot meet the requirement of accurate insulator defect detection for the power line inspection industry.

In 2012, deep learning algorithm became popular, and many studies using deep learning-based object detection methods

achieved performance in power line insulator defect detection. Figure 2 shows the growth trend of insulator defect detection research based on deep learning algorithms from 2010 to 2022. Especially in 2022, there was a huge increase in research work, reaching 2442, which is about the same as the number of articles in 2021. Compared with conventional methods, deep learning-based methods are more flexible and adaptable to insulator defect detection tasks under various backgrounds. Considering that some authors performed the analysis regarding the location of the defect, others only focused on the position of the insulator. Table 1 presents a comparative analysis of what was performed in this field of study from 2019 to 2023, which illustrates selected representative deep learning-based insulator defect methods with several important information in processing stages (w/o data augmentation, insulator location, and detection location).

Generally, to successfully realize insulator defect detection, it needs to obtain the location of insulators and defects. Although some current methods can deal with insulator defects, some urgent issues still need to be solved. In the beginning, the traditional insulator defect detection usually relies on manual patrol to find defects. With the introduction of the concept of smart grid, the UAV patrol was developed to collect insulator data instead of manual patrol. In the insulator defect detection system, there have been some works via traditional methods, such as handcrafted feature RGB colour [11], local binary pattern (LBP) [9], and histograms of oriented gradients (HOG) [27], which have been applied to insulator defect detection [23–28]. Due to the complex outdoor environment, rain, snow, fog, and other weather interference, the detection is largely disturbed. These traditional handcrafted features are sensitive to shape and light variations, and even complex backgrounds. Meanwhile, under the impetus of computer power, the application of deep convolutional networks in object detections becomes more and more popular. Wu et al. [19] proposed an improved you only look once (YOLOv3) method for insulator detection. Guo et al. [21] used a network to localize insulators. In [8, 33–35], these add new convolutional networks to increase accuracy of

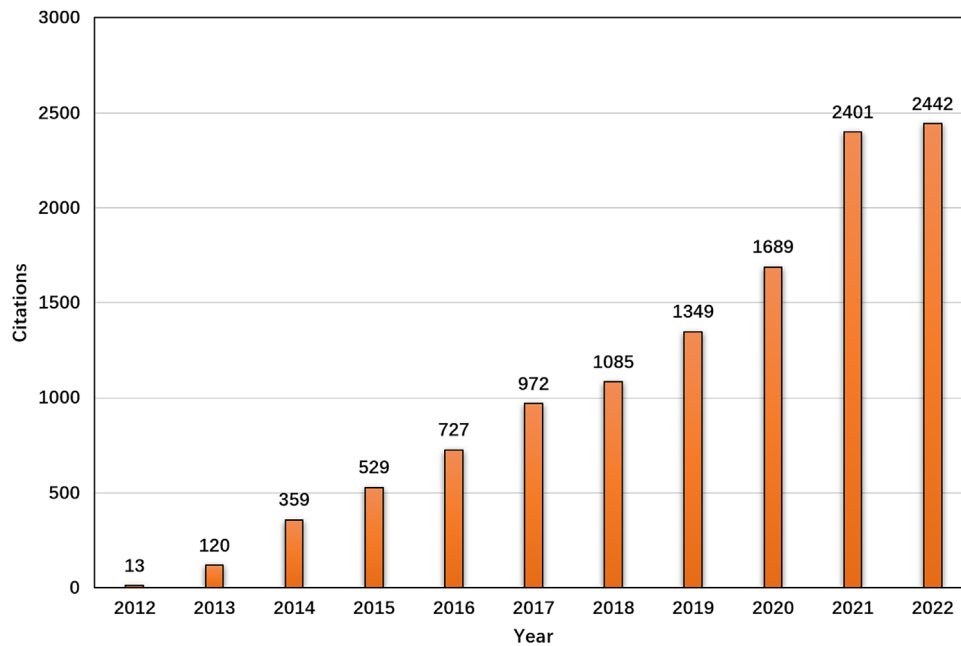


FIGURE 2 The growth trend of insulator defect detection research based on deep learning algorithms from 2010 to 2022 in the Web of Science database.

TABLE 1 Related works about insulator defect detection based on deep learning algorithm.

Author	Year	Main method	Data aug- mentation	Insulator location	Defect location
Sampedro et al. [8]	2019	CNN	✓	✓	✓
Jiang et al. [54]	2019	SSD	✓	✓	✓
Zhang et al. [55]	2019	SWT	✓	–	✓
Miao et al. [62]	2019	SSD	✓	✓	–
Wang et al. [63]	2019	SCA	✓	–	✓
Han et al. [59]	2019	CNN	✓	✓	✓
Tao et al. [30]	2020	CNN	✓	✓	✓
Li et al. [7]	2020	Faster R-CNN	✓	✓	✓
Zhao et al. [57]	2020	AVSCNet	–	–	✓
Wang et al. [22]	2020	ResNeSt	✓	✓	✓
Li et al. [58]	2020	Faster R-CNN	✓	–	✓
Liu et al. [61]	2020	RetinaNet	–	✓	✓
Liu et al. [60]	2021	YOLOv3	✓	✓	–
Wen et al. [56]	2021	Exact R-CNN	✓	✓	✓
Zhao et al. [10]	2021	Faster R-CNN	–	✓	✓
Zhang et al. [37]	2022	Yolov5	✓	✓	✓
Lu et al. [38]	2022	SCRDet	✓	✓	✓
Yang et al. [42]	2022	YOLOv3	✓	✓	✓
Dai et al. [46]	2022	YOLOX	✓	✓	✓
Souza et al. [41]	2023	Hybrid-YOLO	✓	✓	✓

Abbreviations: AVSCNet, automatic visual shape clustering network; CNN, convolutional neural network; R-CNN, region-convolutional neural network; SCA, segment connection algorithm; SWT, square wave transformation; SSD, single shot multibox detector; SCRDet, small, cluttered and rotated objects; YOLOv3, you only look once.

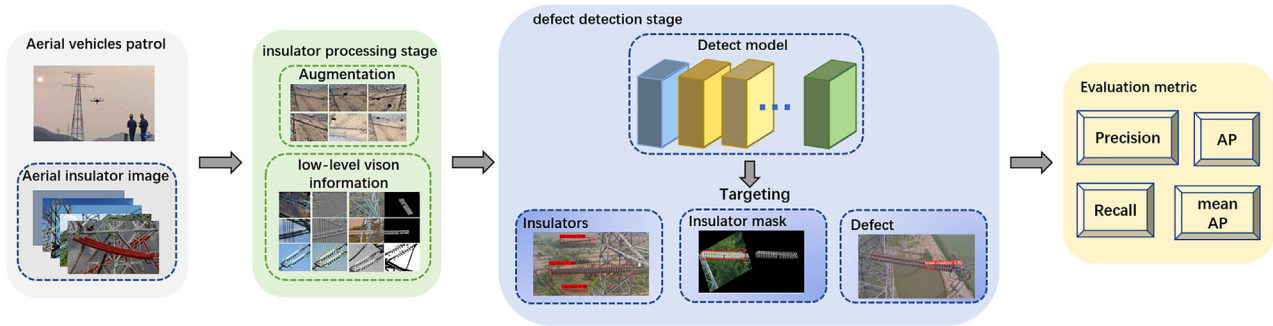


FIGURE 3 The insulator defect detection system. The insulator image is obtained by UAV patrol, then insulator preprocessing stage is used for preliminary image analysis, the defect detection stage to get the target, and finally the evaluation metric is used for evaluation. UAV, unmanned aerial vehicles.

insulator defect detection. However, the challenges, such as small and long insulators, and dense arrangements, always exist outdoors. The objective of this paper is to estimate with the research status of insulator defect detection methods, and to identify its further potential. Specifically, this paper reviews the research methods of insulator defect detection in recent years. At present, these detection methods are not accurate enough in insulator defect location, and insulator defect detection is still in the development stage. Moreover, the authors introduced the overview of image processing algorithm. This paper highlights the key research issues in insulator defect detection, like (i) the problem of insulator defect detection in complex scenarios; (ii) the problem of insufficient small-scale (insulator defect) insulator data; (iii) the problem of insufficient small-scale (insulator defect) insulator data. It also intends to identify research activities, trends, issues, and challenges. The more researchers know about the insulator defect detection, the better they will understand its value. In this case, the research of insulator defect detection is very significant for the safe operation of power system.

The related survey articles [6, 64] provided a classification of insulator defect detection based on the deep learning method. Liu et al. [64] reviewed on online insulator defect diagnosis according to existing research difficulties in application scenarios, like real-time speed, complex background, small-scale object etc. Liu et al. and Liu et al. [6] reviewed on automatic fault diagnosis in power line inspection according to the object detection diagnosis procedure, different from other related survey articles [6, 64]; the proposed survey shows a novel respective of insulator defects taxonomy according to processing stage in deep learning framework, which divides into two-stage procedure and one-stage procedure. As shown in Figure 3, the proposed survey divides deep learning-based methods into insulator processing stage methods and defect detection stage methods according to different processing stages. The insulator processing stage methods exploit the specific image pre-processing algorithm for data augmentation and low-level vision information extraction. The defect detection stage model can locate and classify diagnosis fault with different task targets, like sequential task strategy and multi-task strategy.

The remaining sections are organized as follows. Section 2 mainly introduces several commonly used image preprocessing methods. Combined with the characteristics of insulator defect and aerial image data, the original image is processed with data preprocessing operations such as weight removal, sampling reduction, image enhancement, and noise removal, which ensure the accuracy and validity of the basic data. Section 3 summarizes the framework of insulator defect detection methods in recent years, and according to the model training process, it can be divided into two types: Multi-task strategy and Sequential task strategy. Section 4 describes the available data sets for insulator defect detection in existing work, and gives the common evaluation criteria for insulator defect detection. Section 5 summarizes the introduction of the current popular insulator defect detection methods and the comparison of experimental results, and analyzes the advantages and disadvantages of each method and its application prospects. Finally, several interesting research directions and trends are put forward in Section 6, like (1) the weakly supervision defect detection method provides positive way to decrease the training label cost; (2) large-scale insulator datasets could improve the detection accuracy by collecting more high-quality training images; (3) a robust defect detection model is necessary for complex application scenarios, which could be inspired by domain transfer field; (4) interpretable fault diagnosis plays an important role in the deep learning-based model and online diagnosis monitoring, which lacks related research nowadays.

2 | INSULATOR PREPROCESSING STAGE

The UAV patrol system transmits high-quality aerial insulator images to the monitoring centre for insulator analysis. The insulator detection model always requires a large amount of computing cost. Besides, the large-scale raw images make it easy to cause information redundancy in the following tasks. Therefore, it is necessary to process original images before insulator defect detection, which aims to eliminate irrelevant information and capture useful information for downstream insulator analysis (like insulator detection, segmentation, recognition, and so

TABLE 2 Insulator preprocessing stage.

Type	Purpose	Method
Data augmentation-based preprocessing	Enhance data	1. Geometric transformation (Flip, Rotate, Crop, Scale, Shift, Random Crop or Pad etc.) [3–7] 2. Pixel transformation Add salt and pepper noise, Gaussian noise, adjust brightness, adjust white balance etc. [3, 5, 7]
Low-level vision information-based preprocessing	Assign labels based on attributes of pixels	HSV colour, RGB colour, HSI channels, LBP etc. [8–11]

Abbreviations: HSI, hyperspectral image; HSV, hue, saturation, and value; LBP, local binary pattern.

on). Most recent research finds suitable image processing methods could help enhance insulator defect detection performance. Here, we further classify former insulator image processing methods into two categories: data augmentation-based preprocessing and low-level vision information-based preprocessing, as shown in Table 2.

2.1 | Data augmentation-based preprocessing

In deep learning model training, large-scale label data is needed to eliminate the overfitting problem. Generally, the more the samples, the better the model performance and the stronger the generalization ability. However, due to the limitation of image capture device, the amount of insulator data is not enough to train a satisfactory depth model. Therefore, a large number of researches have artificially enhanced sample data to increase the number of training sets. Liu et al. [3] used noise addition and image rotation to change the original image style. Wei et al. [4] expanded the image of insulator defects by flipping, translating, rotating, and more. Li et al. [5] used rotation, mirror flip, contrast, brightness adjustment, and noisy pixel methods to augment insulator images. Liu et al. [6] used augment data by cropping, flipping, and colour jittering against overfitting. Li et al. [7] used undergo mirroring, rotation, affine transformation, Gaussian white noise, brightness and colour transformation, and other data augment operations. Zhang et al. [37] used random brightness and fog thickness to enhance the insulator dataset, constructed a new synthetic foggy insulator dataset (SFID). And the augmentation of foggy data makes the model more robust in both sunny and foggy scenarios for detecting insulator defects.

Insulator geometric transformation is a commonly used data augmentation method in insulator defect detection. And we summarized the role of data augmentation as follows:

- 1) *Avoid over-fitting.* To make sure the training set containing diverse domains data, several image processing and generation methods are utilized to enrich data types and further avoid over-fitting problem.
- 2) *Improve the robustness of the model.* The real-world scenarios are always complex where the trained model is vulnera-

ble to the attack. It is necessary to learn adversarial noise and masks to improve its robustness.

- 3) *Increase training scale of data* to improve the generalization ability of the model.
- 4) *Avoid sample imbalance.* It is easy to collect extremely unbalanced positive and negative samples of insulator in the insulator defect detection task. Thus, the data augmentation strategy can help improve the number of small-scale samples to avoid unbalance problem.

2.2 | Low-level vision information-based preprocessing

Common materials for insulator strings have unique colour classifications, such as glass insulators are green, and composite insulators are red, which make them easy to be observed outdoors. Therefore, researchers used colour determination to identify insulators. Sampedro et al. [8] applied a gamma correction technique to the original image by applying a look-up table (LUT) to find the corresponding colour gamut at the insulator position for pixel enhancement. Huang et al. [9] converted the RGB colour model of the aviation insulator image into the HSI space model, which was used as the input image of the convolutional neural network (CNN) to increase the ability to distinguish various image features, and then used LBP to extract the texture features of aerial images connected in a CNN, thereby improving the classification performance of the CNN. Zhao et al. [10] obtained the image that was clipped by adaptive threshold based on Hue, Saturation, and Value (HSV) colour space and horizontally corrected and then calculated the pixel. Zhai et al. [11] utilized the unique distribution of glass insulators and ceramic insulators in the RGB colour space to identify insulators from the background by setting the corresponding thresholds for the R (red), G (green), and B (blue) colour components. Huang et al. [9] used LBP to extract the texture features of aerial images connected in a CNN. Tao et al. [12] used four methods for image enhancement. An affine transformation is initially applied to the image, changing the size, orientation, and position of the insulators in the original image. The U-Net [14] algorithm based on TV-based [13] is then used to separate the insulator from the original background and fuse it with the new background. Finally, Gaussian

blur and brightness transformation are used to change the blurriness and brightness of the image to simulate the real imaging environment.

After the input of digital images, the image processing system usually treats all images as 2D signals, enhances relevant information, and eliminates irrelevant information, so as to improve the reliability of feature extraction, matching, and recognition. Image processing is a method of enhancing the quality of an image or gathering alerting insights from an image and feeding it to an algorithm to predict later things. Therefore, researchers find semantic segmentation to be a suitable way to preprocess insulator images for defect detection. Image segmentation is the key process of dividing insulator image into several specific regions with unique properties and inherent objects. Li et al. [7] utilized the U-Net network to divide normal and defective pixels of insulators, which are extracted from high-resolution raw images for semantic segmentation. Zhao et al. [10] obtained the image that was clipped by adaptive threshold based on HSV colour space and horizontally corrected and then calculated the pixel. Zhai et al. [11] used the morphological algorithm [15] to highlight the position of the missing part to locate insulator defects. Due to the unique shape and colour characteristics of insulators, most researchers always designed edge-based low-level information-based preprocessing to obtain the processed insulator images with more salient information for downstream tasks. Furthermore, many researchers combine insulator low-level vision information-based preprocessing to improve the performance of insulator defect detection.

The complex outdoor landform environment makes the mentioned low-level vision information-based preprocessing methods only applicable to the specific environment where the colour of insulators is significantly different from the surrounding. Especially, the colour channel processing method will fail when the glass insulator (with green colour) is in the forest and woods.

3 | DETECTION NETWORK ARCHITECTURE

Insulators are a significant component of transmission lines, the only electrical insulation material, and also an important structural support. Various mechanical and electrical stress insulation failures caused by changes in the environment and electrical load conditions will shorten the operation period of transmission lines. Therefore, regular inspection of insulators is the main means to ensure the safe and stable operation of transmission lines. Transmission line insulators mainly use disc suspension glass insulators and rod suspension composite insulators.

In this paper, we divide the existing insulator defect detection methods into two categories. The insulator defect detection methods are innovatively classified according to the steps of the model task. One is to locate the insulator and its defect directly on the input insulator image through the target detection algorithm, named the multi-task strategy. The second type is to first locate the insulator from the input image and then

TABLE 3 Defect detection stage.

Type	Purpose	Method
Multi-task strategy	Using a network model export insulator detection and defect identification.	—
Sequential task strategy	Step 1: insulator location (insulator detection involves getting the location of the insulators from the collected insulator dataset)	Detection-based sequential task
	Step 2: insulator defect location.	Segmentation-based sequential task

locate the defect on the positioned insulator image, which is called sequential task strategy, as shown in Table 3.

Compared with the multi-task strategy, the sequential task strategy outputs the insulator position first, and then the insulator position is used for defect identification, which reduces the calculation amount of a small target defect detection network, reduces the memory consumption, makes the calculation speed faster, and improves the accuracy of defect detection.

3.1 | Multi-task strategy detection network

In this section, we introduce the multi-task strategy detection network in detail, and show the illustration of insulator detection in Figure 4. The advantages of these kinds of methods are fusing more information and reducing computing cost.

These kinds of insulator detection methods usually use deep learning networks such as Region-CNN (R-CNN), Faster R-CNN, and YOLOv1-v7 to directly perform insulator detection and defect identification in the original image [7, 16–18]. The others use the above networks as the main network for module optimization to achieve insulator detection and defect identification, thereby improving detection efficiency [19–21]. In [16], the faster RCNN network was used for insulator detection and achieved good recognition results, the accuracy reached 90.5%. In [17], the insulator detection and defect identification were detected respectively by Faster R-CNN, achieving a precision of 94%. However, the effect is not ideal when there is an obscuration between the insulators. In [18], they utilized YOLOv3 to detect the insulator detection and defect identification in the original image. In [7], using the YOLOv5 detection method to obtain the highest average precision (AP) value, the insulator detection AP reached 96.47%, and the defect identification AP reached 99.17%. Compared with YOLOv3, the accuracy and efficiency of defect recognition are significantly improved. It makes up for the inaccuracy of YOLOv3 in small target recognition. Although the original objection detection network has a good effect on insulator defect detection, its network structure is huge, which requires huge computing power and takes up a lot of memory. In order to reduce the computational cost, Wu et al. [19] proposed an improved YOLOv3 method for insulator detection, which speeds up the calculation speed by improving the backbone feature extraction network into a lightweight mobile net network. However, the lightweight network mainly uses the following sampling methods to extract

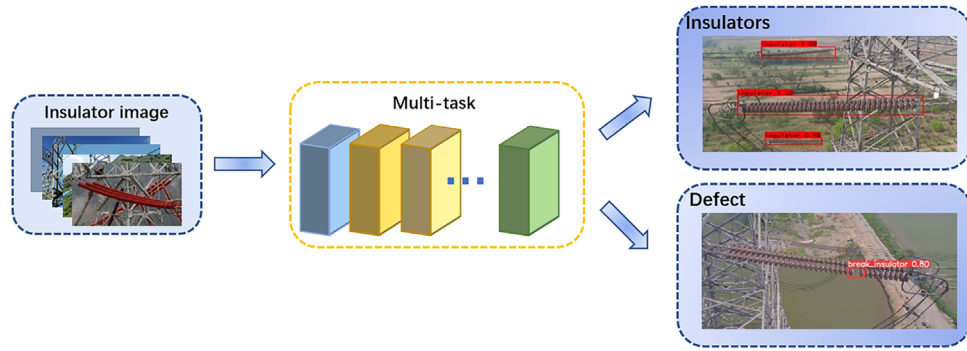


FIGURE 4 The multi-task strategy of insulator defect detection. The insulator image is transmitted to the detection network, and the specific position of the insulator and the position of the insulator defect are output for insulator defect detection.

features, and the channel increase is not enough, resulting in a serious loss of high-level semantic information, further reducing the accuracy. Wu et al. [20] designed an insulator defect detection method based on region of interest (ROI) by improving YOLOv3. Although this method does improve the quality of insulator defect detection in most aerial images, it is still not a real-time solution. Guo et al.'s [21] Faster R-CNN, which combined the deep residual networks and soft-non-maximum suppression (NMS), detected insulators and defects simultaneously in the original image. Wang et al. [22] added an improved region proposal network (RPN) to ResNet for feature extraction to better detect tiny defects on insulators. Liu et al. [3] designed an insulator detection network MTI-YOLO (YOLO-tiny for insulator) for complex aerial images, using the spatial pyramid pooling (SPP) in the detection headers to improve the detection accuracy and the feature expression of specific sizes of insulator. When the colour of the background is similar to that of the insulator, its detection efficiency is not ideal.

The multi-task strategy in the above insulator defect detection methods can directly obtain detection results with fast operation speed, especially the successful detection results based on the YOLO model. However, the detection principle and process based on the defect detection strategy model are unfriendly to the small target detection criteria, and it is easy to miss detection when detecting small defect targets. In addition, these kinds of methods are possible to fail when there is an obscuration [17] between the insulators.

3.2 | Sequential task strategy detection methods

The high-resolution aerial image is scaled to a fixed size, which causes some information about the original image to be lost. The feature map obtained through the subsampling operation make it easy to ignore tiny defective objects. Therefore, many literatures use a Sequential task strategy which first locates the insulator and then conducts defect detection on the insulator. The advantages of this kind of methods are reducing the background interference of small target detection and computation and memory consumption. Besides, the Sequential task strategy could further accelerate the detection speed and improve accu-

racies of defect detection. The illustration of the Sequential task insulator defect detection network is shown in Figure 5.

Here, we classify the Sequential task strategy insulator detection network into two different categories: Detection-based sequential task and Segmentation-based sequential task. The Sequential task strategy is differentiated according to the task content of the first step of the insulator defect detection model. When the insulator defect detection model of the Sequential task strategy outputs the insulator position in the first step, it is named as the Detection-based sequential task. When the output of the first step is insulator mask, it is named Segmentation-based sequential task.

3.2.1 | Detection-based sequential task

With the development of machine learning, more and more researchers try to apply detection-based sequential task methods to solve the object detection task. For example, Elsis et al. [51] utilized the YOLOv3 algorithm to count person numbers first to help enhance decision about energy consumption effectively. They used this idea of detection-based sequential task to detect objects and improve efficiency. The same way can also be applied to the detection task of insulator defect. Miao et al. [29] proposed an automatic multi-level feature extraction model detection method based on an SSD network and implemented a two-stage procedure fine-tuning strategy for the proposed model training to obtain a suitable model for insulator detection. However, the single background of the aerial images in the test cannot reflect the diversity of real aerial image scenes. Tao et al. [30] proposed a novel deep CNN cascading network that transforms defect identification into a two-level object detection problem to improve the accuracy of insulator defect detection. First of all, an insulator is located by a trained visual geometry group (VGG)-16 network, and then, a ResNet-101 network is trained to recognize the insulator and its defects. Han et al. [31] proposed a cascaded model for insulator multi-defect detection. The deep learning network is first used to locate the ROI the insulator strings, and then the YOLO network is trained to detect insulator multi-defects. Sadykova et al. [32] realized a real-time method of insulator detection, the YOLO-v2 network for insulator detection, and then adopted CNN to assess

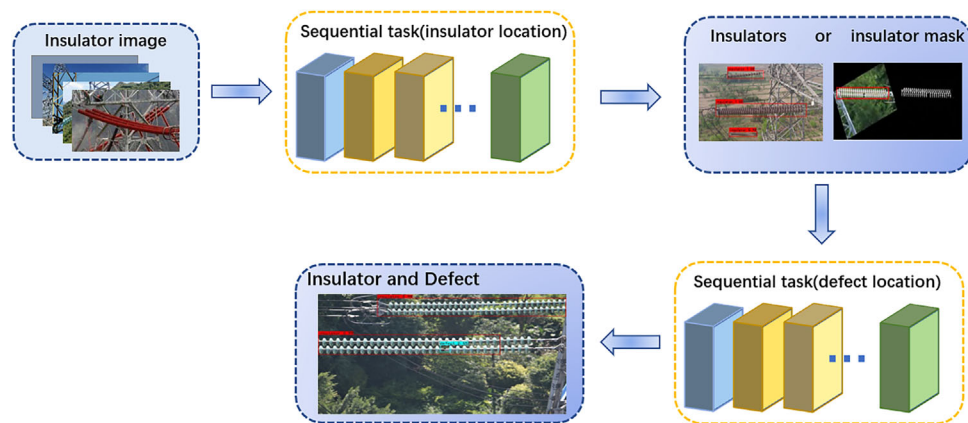


FIGURE 5 The Sequential task strategy of insulator defect detection. The insulator image is transmitted to the first layer detection network to output the specific position of the insulator, and then the image containing the insulator position information is transmitted to the second layer detection network to output the insulator defect position to achieve the purpose of insulator defect detection.

the insulator condition which identified defects through classification. Although these sequential task methods fully integrate global and local information to improve the detection accuracy. However, insulator detection and insulator defect detection are two independent networks, resulting in excessive usage of computing resources. Researchers find detection-based sequential task model could achieve better detection accuracy and computational speed. Sufficient experimental results also prove the detection-based sequential task model would make insulator defect methods more robust in complex surroundings.

However, the detection-based sequential task model brings high computation and large memory consumption. Therefore, in order to optimize the parameters learning strategy, many researchers have proposed the novel designed segmentation-based sequential task insulator detection model, which can effectively solve the high computational load of deep learning network and improve the insulator defect detection accuracy.

3.2.2 | Segmentation-based sequential task

The segmentation-based sequential task insulator defect detection methods require a series of image preprocessing and feature extraction to complete the defect detection. Besides, the complex background and object interferences of aerial insulator image will result in poor robustness of the insulator defect detection model based on segmentation sequential tasks. Therefore, the complete feature extraction module plays a crucial role in the insulator defect detection task. Li et al. [7] proposed a method that cascades detection and segmentation networks to identify the defect; the Faster R-CNN was carried out to capture both defects and insulators in the entire image. Then fed the detect into the U-Net network to further inspect, which reduced missed detection. In [8, 33–35], Faster R-CNN network was used to detect the insulators in the original image, and then the local insulator diagram was cut out according to the coordinates for further semantic segmentation. Ling et al. [33] used U-Net to segment the pixels of the missing insulator defect part, and fully

convolutional networks-stride 8 (FCN-8s) could extract each plate of insulator string for defect identification. Wang et al. [35] and Gao et al. [34] employed FCN-8s to segment the insulator string, and the latter [35] fed the insulator string mask into GoogLeNet to classify normal and defective. Sampedro et al. [8] were similar to it, but they omitted the procedure for detecting insulators, using a U-Net, which added a new skip connection to directly segment the cap and disk in the original images and sent the mask to the CNN classification network for defect identification. However, all the insulators in the results they gave were single strings, which can only identify a variable number of discs, and are not diverse in perspective. To obtain the fine contour of the insulator, Zuo et al. [28] combined the features of the Haar-like features, integral graph features, and Directional Gradient Histogram features to train cascade classifiers and support vector machine (SVM) classifiers that can identify and locate the insulator. Then the defect identification is achieved using pixel analysis of segmented insulators. Zhao et al. [10] proposed an improvement on the Faster RCNN model using feature pyramid networks (FPNs) for insulator localization in complex background images. The target area was segmented using the HSV colour space adaptive thresholding algorithm. In order to reduce the workload of insulator defect labelling researchers, Shi et al. [36] proposed a cap-count guided weakly supervised method, which only required insulator boundary frame labelling. Firstly, cap labelling is generated automatically by giving the boundary frame and cap number of insulators. Then the deep learning model is used to detect the insulator string and cap. Finally, the position of the cap is measured using the geometric constraints of the cap. Lu et al. [38] proposed a three-step outdoor insulator surface defects segmentation method. In this framework, outdoor support insulators are first located using the small, cluttered and rotated objects (SCRDet) network. Second, a resolution-improved insulator reconstruction network (RIIR-Net) is proposed to reconstruct normal insulators. Finally, the defect area is divided according to the colour feature and area, and the residual diagram between the original defect insulator and the reconstructed normal insulator is obtained.

Segmentation-based sequential task methods are usually used for defect identification to reduce network and computational complexity, and deep learning methods are utilized for insulator detection. These methods of deep learning combined with traditional use information both of global and local images to identify insulator defects, reducing missed detections and improved insulator detection performance in complex backgrounds. Although the sequential task strategy approach can improve the detection ability of small targets, it depends on the accuracy of insulator detection or image segmentation. When the result of boundary frame and segmentation mask is not accurate, it will have a bad influence on the subsequent defect detection, and even cause false positive or omission of defects.

This section describes the two types of insulator defect detection methods in detail. It can be seen that the above-mentioned different methods have their advantages and disadvantages. For example, the multi-task strategy detection method has a fast detection speed, but easily leads to information redundancy and high cost. Insulator detection in a complex background is prone to missed detection, especially in the detection of small targets. The sequential task strategy detection method optimizes the network, which improves the detection speed and the detection accuracy of small targets. The experimental results prove this kind of detection methods could be effective even in complex scenes. However, the sequential task strategy detection method is complicated and still requires significant computational resource to train. The most important is that the image segmentation result would affect the accuracy of defect detection.

4 | DATASETS

In the previous sections, we reviewed related field methods' aim to detect insulators and defects. However, most works conduct experiments with personal private datasets. Personal datasets usually have problems such as insufficient insulator defect image data, insufficient insulator types, uneven data distribution, and single structure. Mostly researchers used personal datasets to conduct comparative experiments, but the detection results are not reproducible.

4.1 | Public datasets in insulator defect detection

Due to the special confidentiality of power systems, many power data are not disclosed, so this article has collected some public datasets for insulator defect detection and attached a download from GitHub for easy use.

4.1.1 | The China power line insulator dataset (CPLID)

Tao et al. [30] collected and constructed an insulator public dataset called china power line insulator subset (CPLID), which

can be downloaded from GitHub¹. Some representative insulator images were selected from the CPLID dataset, as shown in Figure 6.

The CPLID dataset provides normal insulator images captured by UAV and synthetic defective insulator images, of which Normal-Insulators are 600 images, and there are 248 images containing insulator defects. This dataset presents various backgrounds of aerial insulator, such as cities, rivers, fields, mountains etc. The defect insulator image is segmented from a small part of the original image by the algorithm in TVSeg. Affine transformation is used to enhance the image, and the enhancement results are many original mask images. Then, the remaining images are divided to make the insulator image attached in different backgrounds.

4.1.2 | The unifying public insulator datasets (UPID)

Andrel et al. [39] realized further data augmentation of CPLID through random affine transformation, Gaussian blur, non-destructive transformation, and other methods, and the unifying public datasets for insulator (UPID) with 6860 training and test images was constructed, which can be downloaded from GitHub², as shown in Figure 7.

4.1.3 | The synthetic foggy insulator dataset (SFID)

Zhang et al. [37] used random brightness and fog thickness to enhance the UPID dataset, constructed a new SFID, which can be downloaded from GitHub³. This dataset contains 13,718 images, as shown in Figure 8.

4.1.4 | The insulator defect image dataset (IDID)

Lewis et al. [40] collected and constructed a dataset of high-resolution photos of insulator chains containing defective components, named insulator defect image dataset (IDID), which can be downloaded from GitHub⁴. The dataset has three subcategories: flashover damaged insulator shells, broken insulator shells, and good insulator shells, as shown in Figure 9. The dataset consists of 1596 images.

4.2 | Experiment

4.2.1 | Dataset protocols

Dataset is a prerequisite for training a good network. If the quality of data set is not guaranteed, it will be difficult to get good

¹ CPLID: <https://github.com/InsulatorData/InsulatorDataSet>

² UPID: <https://github.com/heitorcfelix/public-insulatordatasets>

³ SFID: <https://github.com/zhangzhengde0225/FINet>

⁴ IDID: <https://doi.org/10.21227/vkdw-x769>

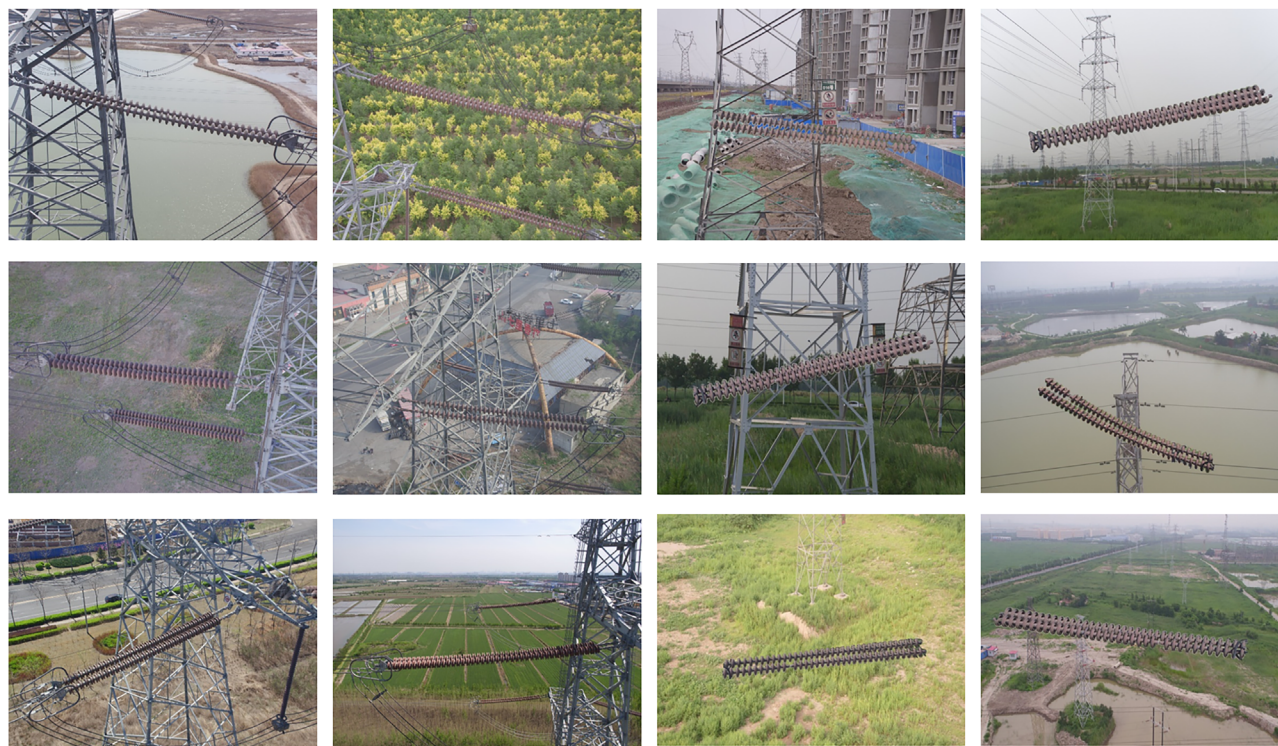


FIGURE 6 Image of public representative insulator detection dataset (CPLID). The two columns on the left are images of normal insulators under different backgrounds in the CPLID dataset, while the two columns on the right are images containing insulator defects under various backgrounds.

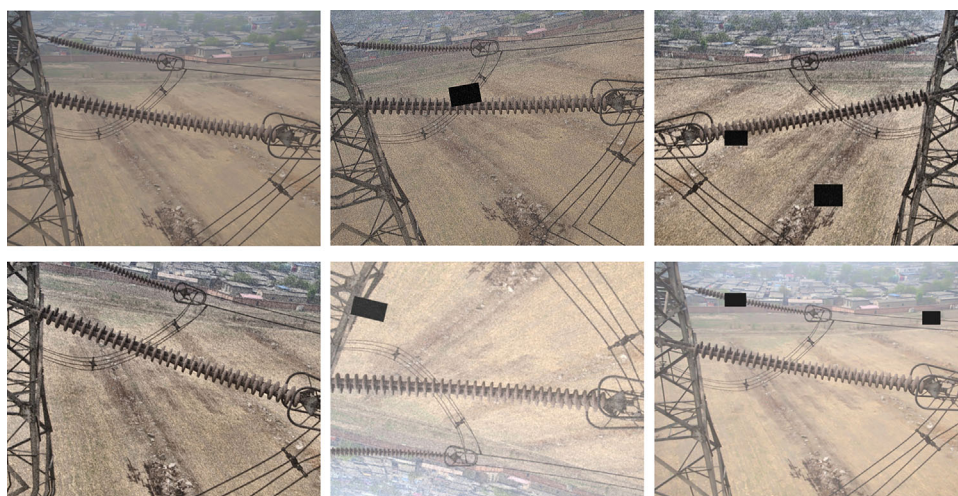


FIGURE 7 Image of unifying public datasets for insulator (UPID). The above figure shows the further data augmentation of CPLID by random affine transformation, Gaussian fuzzy transformation, and lossless transformation. CPLID, China power line insulator dataset.

network parameters. Therefore, the selection and construction of data sets are important. Before the deep learning algorithm is implemented, it is usually necessary to divide the data set, which is usually divided into training set, test set, and partly verification set. The training set is used for the training model, and the parameters of the fitting curve are determined through the data of the training set. Validation set is used for model selection, final optimization, and determination of the model, as well as to

assist model construction. The test set is to test the accuracy of the trained model, judge the discriminant ability of the learner to the new sample, and test the error. The amount of training set data generally accounts for 2/3 to 4/5. In practice, the ratio of training set data to test set data can be 6:4, 7:3, or 8:2 based on the size of the entire data set. You can use 9:1 for large amounts of data. The specific method varies according to the data set.



FIGURE 8 Image of synthetic foggy insulator dataset (SFID). The above figure shows the further data augmentation of UPID by random brightness and fog thickness. UPID, unifying public insulator datasets.



FIGURE 9 Images of insulator from the insulator defect image dataset (IDID). The dataset contains insulator pictures in various colours, diverse positions, and different backgrounds.

In work [30], the CPLID dataset is used for insulator defect defection. In total, 1186 images in the experiment dataset for training and 770 images for testing were carried out insulator location; 792 images for training and 264 images for testing were carried out defect detection (location).

In work [37], the UPID data set used for the fog free test, which contained 1339 insulators and 371 self-detonation defects. The SFID data set is a fog containing test set containing 2743 insulators and 760 self-detonation defects. These models are trained in a finite number of 100 epochs.

In work [46], the IDID data set which contains 6860 and 171 images of insulators for training and testing. The training set includes 4880 images of insulators in normal status and 1980 images of defective insulators, while the test set comprises 121 image instances of good insulators and 50 examples with insulators faults.

4.2.2 | Evaluation metric

In the evaluation criteria of insulator defect detection, several metrics widely used in object detection are usually used for evaluation: precision (P), recall (R), average precision (AP), and mean AP (mAP). The definitions of each indicator are given as follows:

$$P = \frac{T_p}{T_p + F_p} \quad (1)$$

$$R = \frac{T_p}{T_p + F_N} \quad (2)$$

$$AP = \int_0^1 P(R) dR \quad (3)$$

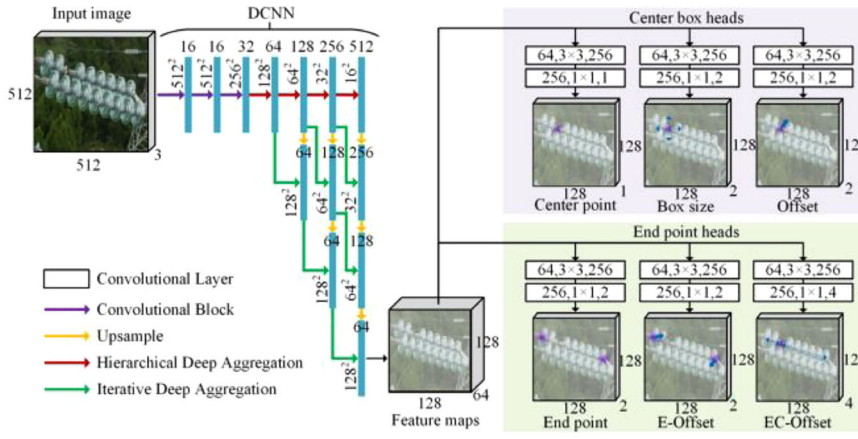


FIGURE 10 Flowchart of the Box-Point Detector method for insulator defect detection. Given 512×512 resolution detection image, DCNN is used to extract high quality feature map, and then down-sampled to generate high resolution 128×128 feature map. More original fault information is retained, which improves the diagnosis capability of small faults. DCNN, deep convolutional neural network.

$$mAP = \frac{\sum_{i=1}^N AP(i)}{n} \quad (4)$$

where T_p is the correct detection result, F_p is the wrong detection result, and F_N is the wrong undetected result. Precision is the indicator for evaluating the accuracy of the prediction result, while recall is the indicator for evaluating the incomplete prediction result. The AP value is the integral of the precision value at different recall rates and is used to measure how good the trained model is in each class. mAP is the average of AP values for different classes.

5 | EXPERIMENTAL RESULTS AND ANALYSIS

This paper selects representative methods from Multi-task strategy and Sequential task strategy for comparison. Subsequently, we analyzed the experimental results of various representative methods, and the results of each evaluation index were reported in the original literature. The details and methods are shown in Table 4.

X. Liu et al. [6] proposed a novel Box-Point Detector for insulator defect detection. The introduction of the model is shown in Figure 10.

The method consists of a deep convolutional neural network (DCNN) and two specially designed convolutional head branches. Firstly, the high-quality features of aerial images are extracted by DCNN, and then two branches are input. The central box head is responsible for the insulator fault information, and the endpoint box head corresponds to the insulator information, which is equivalent to the context information near the fault. Both groups share a single DCNN; they use the loss function as follows:

$$L_{cl} = \frac{-1}{N} \sum_{xye} \begin{cases} (1 - Y_{xye})^\alpha \log(Y_{xye}) \text{ if } \widehat{Y}_{xye} = 1 \\ (1 - \widehat{Y}_{xye})^\beta Y_{xye}^\alpha \log(1 - Y_{xye}) \text{ otherwise} \end{cases}, \quad (5)$$

$$L_{off} = \frac{1}{N} \sum_{\hat{p}} |O_{\hat{p}} - o| \quad (6)$$

$$L_{size} = \frac{1}{N} \sum_{\hat{p}} |S_{\hat{p}} - s| \quad (7)$$

where α and β are hyper-parameters, N refers to the number of centre points in the input image, $O_{\hat{p}}$ is the predicted offset in the local offset feature-map. L_{cl} is to train the head of centre point heatmap, L_{off} is the loss to train the head of local offset feature-map, and L_{size} is to train the head of box size feature-map.

The final loss function is the linear combination of former loss terms. The proposed method solves the problem of double calculation, preserves more original fault information by using high-resolution feature maps, and improves the diagnostic ability of small-scale faults ($AP = 0.74$). However, false detection can occur when the insulator string is in the cluttered background and cover of the tower.

Among the YOLO family algorithms for insulation defect detection, YOLOv4 has the lowest detection accuracy for transmission line insulator defects, which is mainly reflected in its poor detection effect for small targets such as insulator defects. Li et al. [5] proposed the four types of YOLOv5 series (YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x) for insulator defect detection. The experimental results show that YOLOv5 series are significantly higher than YOLOv4 and YOLOv3, especially YOLOv5m, with map value up to 98%, which has a higher advantage in insulator defect detection. In the YOLOv5 series model, it is found that the accuracy of the model decreases with the increase of the complexity of the model. To a certain extent, due to the uneven data distribution of insulators and insulator damage targets in the data set, the data volume is insufficient and the performance of the model cannot be improved. Souza et al. [41] proposed a Hybrid-YOLO method for the identification of various conditions of insulators. The method combines an optimal YOLO model for insulator detection with an optimal CNN structure for insulator condition classification to produce a model with best performance. Stefenon et al. [46] presented YOLO-drone (YOLOD) to

TABLE 4 Experimental details of representative methods.

Type	Model	Dataset	Year	Evaluation criteria			
				Precision	recall	AP/AP ₅₀ /AP ₇₅ (%)	mAP
Multi-task strategy	Box-point detector [6]	Personal dataset (private)	2021	–	–	73.7/94.0/86.4	–
	YOLOv3 [42]	CPLID (public)	2022	90	91	–	–
	Improved YOLOv3 [42]	CPLID (public)	2022	94	92	–	–
	YOLOv4 [5]	CPLID (public)	2022	–	–	93.84/-/-	94.39
	Yolov5 series [5]	CPLID (public)	2022	–	–	95.67/-/-	95.31
	YOLOv5m	CCIN-detection dataset (private)	2021	95	91	89.72/-/-	–
	YOLOv5l	CPLID (public)	2022	99.4	93.19	–	99.26
	YOLOv5x	IDID (public)	2023	–	–	73.9/98.0/84.6	–
	MTL-YOLO [42]	Personal dataset (private)	2018	96	–	–	–
	Hybrid-YOLO [41]	CPLID (public)	2020	91	96	–	–
Sequential task strategy	YOLOD [46]	CPLID (public) and Personal dataset (private)	2021	–	–	–	91.7 (glass insulators) and 0.92 (composite insulators)
	Morphological algorithm [25]	GCAP dataset (private)	2021	92.86	86.67	–	–
	Cascading CNN [12]	FIN dataset (private)	2022	92.08	90.45	–	–
	Improved faster R-CNN [10]	Self-collected dataset (private)	2022	74	75	–	98
	Cap-count guided weakly supervise [36]	–	–	–	–	–	–

Abbreviations: CNN, convolutional neural network; FIN, four types of insulators; GCAP, glass caps; R-CNN, region-convolutional neural network; RIIR-Net, resolution-improved insulator reconstruction network; YOLOD, you only look only drone; YOLOv3, you only look once.

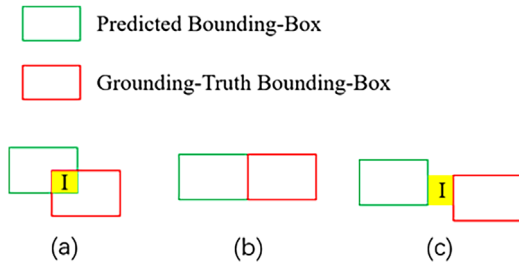


FIGURE 11 The different detection results. (a) Predicted bounding box has an intersection with grounding-truth bounding box. (b) Edge of the predicted bounding box coincides with grounding-truth bounding box. (c) Predicted bounding box has no intersection with grounding-truth bounding box.

address the uncertainty issue by applying a Gaussian prior to the detection heads of YOLOX, the current state-of-the-art model of compact object detectors, for both bounding box regression and corresponding uncertainty estimation. The emergence of the you only look only drone (YOLOD) model solves the problem that the current detection methods cannot predict and deal with the uncertainty of insulator detection. This method refines the boundary box prediction and improves the robustness of detection.

The MTI-YOLO [3] method introduces multi-scale feature detection probe, multi-scale feature fusion structure, and spatial pyramid pool model into the YOLO-tiny network to optimize the detection accuracy, running time, and storage space. The experimental results show that compared with the YOLO-tiny, YOLOv2, and YOLOv3 networks, the proposed method improves the insulator detection accuracy in complex scenarios, and has obvious advantages in terms of running time and memory consumption. Similarly, Yang et al. [42] proposed an improved YOLOv3 network to solve the problem of low insulator defect detection accuracy in UAV detection. In order to improve the detection accuracy of small targets, the FPN of improved YOLOv3 is changed to bidirectional fusion to improve the receiving field of small targets. In addition, Efficient Intersection over Union (EIoU) and Smooth-EIoU loss are added so that the predicted frame almost coincides with the actual frame and the loss converges faster in the training phase. The explanation of EIoU is shown in Figure 11.

The calculation results of Intersection over Union (IoU) and EIoU are shown in Figures 11(a) and (b). As shown in Figure 11 (c), the IoU result is 0 while the EIoU result is negative.

The specific calculation methods of IoU Loss and Smooth-EIoU Loss are given as follows:

$$IoU\ Loss = 1 - \frac{I}{U} \quad (8)$$

$$Smooth_EIoU\ Loss = \left(1 - \frac{I}{U}\right)^2 \quad (9)$$

IoU Loss is the calculation of the intersection between the predicted frame and the actual frame. If the two boxes coincide,

the intersection ratio is equal to 1 and the loss is 0. It means that the degree of coincidence is very high. Regardless of the size of prediction frame and annotation frame, the output IoU Loss is always between 0 and 1, which can truly reflect the detection effect. However, when the prediction box and the label box do not intersect, the distance between the two boxes cannot be reflected. The loss is still 0, and the model cannot be further training. Compared with IOU Loss, Smooth-EIoU loss solves the problem of unbalance of difficult and easy samples by calculating width–height difference respectively instead of aspect ratio, thereby improving the accuracy.

Liu et al. [3] and Yang et al. [42] added model receptive field based on the YOLOv3 model to optimize the detection of small targets. Particularly, Yang et al. [42] also added a new loss function, which made the predicted frame almost coincide with the actual frame, making the model converge faster in the training stage and improving the detection accuracy of the model. In deep learning model, configuring loss functions is one of the most important steps in ensuring that the model works as expected. Therefore, we can consider starting from the structure of the model itself to optimize the detection effect, which can make the model as light as possible but also improve the detection accuracy.

The segmentation-based sequential task methods usually adopt the handcrafted features, such as SIFT [43], HOG [44], and LBP [45] to obtain the position of the insulator string first, and then extract the insulator contour combined with the image preprocessing method. Finally, the insulator defect detection is realized by comparing the normal insulator string. Zhai et al. [25] proposed an algorithmic solution for insulator defect detection that considered the unique spatial and colour features of glass insulators and ceramic insulators.

First, the insulators with different angles can be processed using the Hough Transform to detect the straight lines, and the insulators can be located. Then, the located insulator is processed through morphology, and the spatial features of defect location are analyzed. Finally, according to these rules, the defect location coordinates of insulators are determined. Although this method is suitable for insulator detection of the two types, it lacks universality and robustness to test by handcrafted features, and these rules only apply for insulator defect detection with simple background, no occlusion, and clear profile.

Tao et al. [12] proposed a novel deep CNN cascading architecture for both localization and defect detection of insulators. Figure 12 shows the entire structure of this model.

To train the RPN, the loss function can be defined as

$$L(\{p_i\}, \{t_i\}) = \frac{1}{N_{cls}} \sum_i L_{cls}(p_i, p_i^*) + \lambda \frac{1}{N_{reg}} \sum_i p_i^* L_{reg}(t_i, t_i^*) \quad (10)$$

where i is the object proposal index, p_i is truth label, p_i^* is the proposal object, and if the proposal is for an object, $p_i^* = 1$, but otherwise if $p_i^* = 0$. t_i is the detection box, and t_i^* is the labelled detection box. L_{cls} and L_{reg} are the logarithmic loss and regression loss functions, respectively, and N_{cls} and N_{reg} are the number of classes and detected boxes, and with λ , these three parameters are used to normalize the two loss functions.

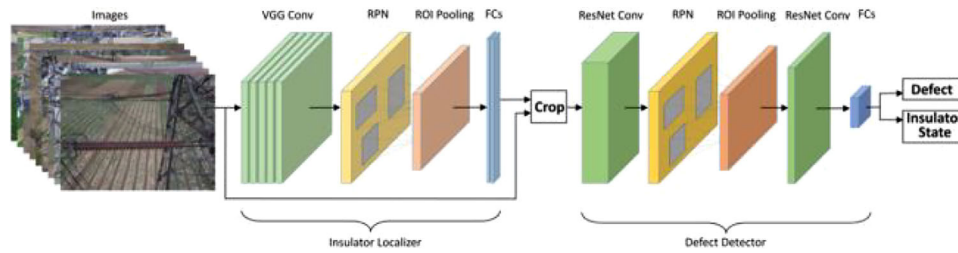


FIGURE 12 Flowchart of the Cascading CNN method for insulator defect detection. The overall detection framework is divided into two parts. The first part locates the insulator, realizes the defect segmentation in the input next part, and finally outputs the insulator and fault location. CNN, convolutional neural network.

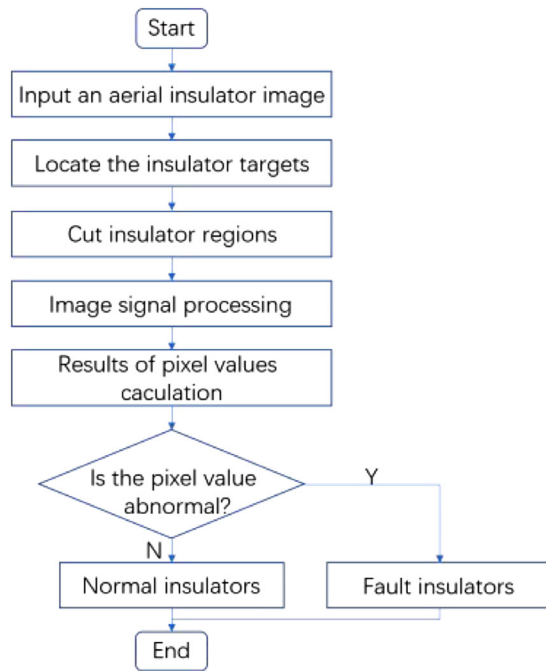


FIGURE 13 Flowchart of the improved Faster RCNN method for insulator defect detection. RCNN, region convolutional neural network.

In this model, they transform defect detection to a two-level detection problem. In the first step, the object detection method is directly used to locate the insulator position, and then the object detection method is used again to locate the defect position on the insulator, fully utilizing the regression performance of CNN. Although the experimental results show that the detection accuracy of insulator defects is higher than that of other detection methods, the biggest problem is that the two-level detection network requires a large amount of memory consumption and takes a long time to calculate.

In a complex background, Zhao et al. [10] proposed the improved Faster RCNN model as an insulator detection method which has obvious advantages compared with the traditional method. The flowchart of insulator fault classification is depicted in Figure 13.

The loss function of RPN is the same as in [12]

$$L(\{p_i\}, \{t_i\}) = \frac{1}{N_{ds}} \sum_i L_{cls}(p_i, p_i^*) + \lambda \frac{1}{N_{reg}} \sum_i p_i^* L_{reg}(t_i, t_i^*) \quad (11)$$

Based on the Faster RCNN model as an insulator detection network, the insulators in the image can be accurately identified and clip the insulator regions. After processing the images, the defect classification of insulator target is carried out by using adaptive threshold segmentation, line detection, image level correction, and pixel value calculation. Finally, the results are output. However, since the detection accuracy of insulator defects largely depends on the segmentation level of the image, it is not applicable to the case that the target insulator is covered.

6 | CONCLUSION AND FUTURE WORK

The report ‘*Performance of China’s Power System Transition*’⁵ says ‘China’s power grid scale has been steadily expanded. By the end of 2021, the length of China’s transmission lines (≥ 220 kV) reached 843,000 km, with a substation capacity of 4.47 billion kVA and a DC converter capacity of 4700 GW; the length of high-voltage distribution lines reached 1,150,000 km.’ China’s power grid is in a new era of fast, healthy, and sustainable development, especially in the field of transmission and distribution, which has broad and huge economic prospects and development potential. The insulator is a special insulation control device that hangs over power transmission lines. Transmission line insulators are exposed outdoors to the effects of sunlight, rain, climate change and chemicals, and dirt that accumulates on the surface of the insulator increasing the conductivity of the surface, making the location more susceptible to discharge [47]. The surface of the insulators is usually made of ceramic or glass material and can crack from damage. When the surface of an insulator is charred, it loses its ability to insulate. The insulator must be replaced to ensure the normal transmission of the power network [48]. Therefore, it is very important to inspect insulators regularly for defects. The traditional manual inspection method is time-consuming, labour-intensive, and has hidden dangers. In recent years, rapid advances in vision-based UAV technology have provided an ideal solution to replace the manual patrol task of performing dangerous power line insulator inspection. Specifically, drones can easily fly over rivers and mountains, track power towers along power lines, and inspect related accessories. Based on these advantages, over the past few years, many researchers have studied the insulator defect

⁵ Report: https://iea.blob.core.windows.net/assets/d689f6ad-72d9-4936-9c5d-5a8dd00942d1/%2F220921PST3_FANGXiaoSong.pdf

detection based on UAV, and the most popular one is the defect research based on deep learning. However, a deep learning model based on insulator defect detection has an inherent fault that needs a large number data sets for training. Although deep learning-based insulators have achieved great progress recently, there still exist several exciting challenges. In this section, we discuss these unsolved problems to inspire other researcher in the community.

- (1) **Weakly Supervision Defect Detection:** Nowadays, the existing insulator defect algorithms mainly focus on locating and identifying faults with fully supervised training data. Large-amount of raw insulator image data are still not annotated accurately due to high time and resource cost. Thus, how to train these defect detection models only with weakly supervised data is an important problem in real-world scenarios.
- (2) **Large-scale Insulator Datasets:** According to the publicly available datasets analysis in the paper, the existing datasets are relatively small-scale and not normalized, which brings in such difficulties in reimplementation. As a potential solution, newly arranging existing datasets and prove public experimental protocols would help other researchers. Besides, large-scale insulator data should be further collected to improve deep learning ability.
- (3) **Robust Defect Detection Model:** In real-world scenarios, these insulator defect detection model would be applied in complex scenarios, which results in insulator changes in object scale, pose, illumination, occlusion, etc. Domain transfer and generalization algorithm may inspire researchers to explore more robust deep learning models.
- (4) **Interpretable Fault Diagnosis:** Most existing deep learning-based models are still a 'black box', which lack theoretical explanation. More work should explore the interpretability of fault detection model, but not only the better performance. Meta learning and hybrid knowledge model may provide a feasible way to solve the interpretability problem in the future.

AUTHOR CONTRIBUTIONS

Yue Liu: Data curation; Methodology; Writing—original draft. Decheng Liu: Investigation; Methodology; Writing—original draft. Xinbo Huang: Project administration; Writing—review & editing. Chenjing Li: Writing—review & editing.

ACKNOWLEDGEMENTS

This work was supported in part by the Scientific Research Program Funded by Shaanxi Provincial Education Department (Program No. 21JK0661), in part by the Key Research and Development Projects in Shaanxi Province (2021GY-306), in part by the Key R&D plan of Shaanxi (2021GY-320, 2020ZDLGY09-10), in part by the Natural Science Basis Research Plan in Shaanxi Province of China (Program No. 2022JQ-568, No. 2022JQ-696), and in part by the Fundamental

Research Funds for the Central Universities (Program No. XJS221502).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Liu, Y., Liu, D., Huang, X., Li, C.: Insulator defect detection with deep learning: A survey. *IET Gener. Transm. Distrib.* 17, 3541–3558 (2023). <https://doi.org/10.1049/gtd2.12916>